C 22563 **(Pages : 3)** **Name.....**

Reg. No.....

FOURTH SEMESTER M.Sc. DEGREE (CBCSS) [REGULAR/ SUPPLEMENTARY] EXAMINATION, APRIL 2022

Mathematics

MTH 4C 15—ADVANCED FUNCTIONAL ANALYSIS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

General Instructions

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

Part A

Answer **all** *questions.*
Each question carries a weightage 1.

1. Let $A : X \to X$ be any bounded operator. Then show that non-zero eigen vectors corresponding to distinct eigen values are linearly independent.
2. For any compact operator T, show that $0 \in \sigma(T)$.
3. Show that the eigen values of a self-adjoint operator are real.
4. Let E be a linear space, and P be a projection on E. Let $E_1 = Im P$ and $E_2 = Ker P$. Show that E is the direct sum of E_1 and E_2.
5. Show that every orthoprojection P satisfies $0 \leq P \leq I$.
6. State Banach-Steinhaus theorem.
7. Define sub-linear functions on a linear space. Give an example which is not linear.
8. Define Banach algebra and give an example.

(8 × 1 = 8 weightage)

Turn over

Part B

Answer any **two** *questions from each module.*
Each question carries a weightage 2.

Module I

9. Show that for every $\in > 0$, there is only a finite number of linearly independent eigen vectors corresponding to eigen values λ_i with $|\lambda_i| \geq \in$.

10. If A is a symmetric operator, show that $\|A\| = \sup_{x \neq 0} |\langle Ax, x \rangle| / \|x^2\|$.

11. Let H be an infinite dimensional Hilbert space. Then show that :

    i) If a positive eigen value exist, the $\max_{\|x\|=1} \langle Tx, x \rangle = \lambda_1^+$; if a negative eigenvalue exists, then $\min_{\|x\|=1} \langle Tx, x \rangle = \lambda_1^+$.

    ii) $\langle Tx, x \rangle$ for every $x \in H$ if and only if there are no negative eigenvalues.

Module II

12. Prove that for any self-adjoint operator $A \in L(H)$ the residual spectrum is empty.

13. Let A be an operator such that $mI \leq A \leq MI$ for some m, $M \in R$ and let P be a polynomial satisfying $P(z) \geq 0, \forall z \in [m, M]$. Then show that $P(A) \geq 0$.

14. Define Orthoprojection. Also show that a projection is an orthoprojection if and only if ImP $\perp$ KerP.

Module III

15. Let X and Y be Banach spaces. Let $A : X \mapsto Y$ be a closed graph operator and Domy A = X. Then show that A is continuous.

16. For a real Banach space X, show that the unit ball D (X*) is a compact set in the w^*-topology.

17. Show that in a Banach algebra, every proper ideal is contained in a maximal ideal.

(6 × 2 = 12 weightage)

Part C

Answer any **two** *questions.*
Each question carries a weightage 5.

18. Let T be a compact operator and $\lambda \neq 0$. Then show that $\Delta_\lambda = \overline{\Delta}_\lambda$.

19. State and prove the first Hilbert-Schmidt theorem.

20. Explain the construction of the spectral integral.

21. If K is perfectly convex in a Banach space X, then show that $\overset{o}{K} = \overset{c}{K} = \overline{\overset{c}{K}} = \left(\overline{\overset{o}{K}}\right)$.

(2 × 5 = 10 weightage)